Course Code : ECT 352 **KQLR/RS – 24 / 1605**

Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

CONTROL SYSTEMS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory.
(2) All questions carry marks as indicated against them.
(3) Use graph paper wherever necessary.
(4) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) Using the block reduction technique, determine the transfer function C(s)/R1(s) of a system represented by the block shown in Fig. 1.

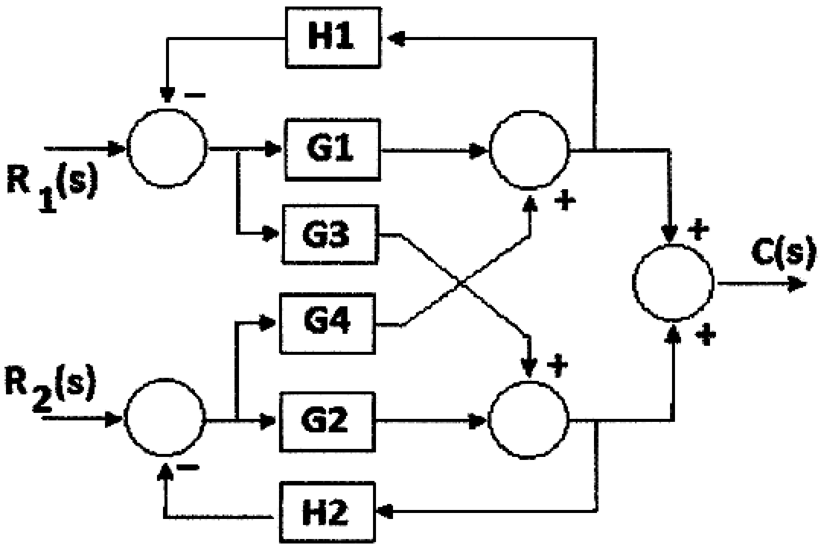

Fig. 1 6(CO2)

(b) Represent the following set of equations by a signal flow graph and determine the overall gain relating to x_5 and x_1 using Mason's gain formula.

$$x_2 = p\,x_1 + q\,x_2 \quad ; \quad x_3 = r\,x_2 + s\,x_4$$
$$x_4 = t\,x_3 + u\,x_5 \quad ; \quad x_5 = v\,x_4 + w\,x_2$$

6(CO2)

2. (a) Obtain an expression for unit step time response of a control system whose transfer function is given by –

$$\frac{C(s)}{R(s)} = \frac{0{\cdot}0168}{s^3 + 0{\cdot}42s^2 + 0{\cdot}17s + 0{\cdot}0168}$$ 6(CO4)

(b) A second order control system is represented by a transfer function as :

$$\frac{\theta_0(s)}{T(s)} = \frac{1}{Js^2 + fs + K}$$

where θ_0 is the proportional output and T is the input torque. A step input of 10 Nm applied to the system and test results are given below :

(a) $M_p = 6\%$,

(b) $t_p = 1$ sec,

(c) The steady state value of the output is 0·5 radian

Determine the values of J, f and K. 6(CO4)

3. (a) Investigate the stability of a control system using Routh's criterion having open-loop transfer function given below :

$$G(s)H(s) = \frac{K(s+2)}{s(s+3)\ (S^2 + 2s + 3)}$$ 6(CO3)

(b) Construct the root locus for a system with open-loop transfer function,

$$G(s)H(s) = \frac{K}{s(s+4)}$$

Addition of pole at $s = -6$ leads to instability of the system, justify. 6(CO3)

4. (a) Sketch the Bode plot for the open-loop transfer function of a unity feedback system –

$$G(s) = \frac{2(s+0{\cdot}25)}{s^2(s+1)\ (s+0{\cdot}5)}$$ 6(CO4)

(b) The open-loop transfer function of a unity feedback system is –

$$G(s) = \frac{K}{s(s+1)\ (s+2)\ (s+4)}$$

Sketch a polar plot and determine the intersection point; if any, with the real and imaginary axis. 6(CO3)

5. (a) Identify the state variables required to obtain phase variable state for the RLC series electrical network. Also find the state model for the RLC series electrical network. 6(CO5)

(b) Discuss the following common non-linear system behaviors :

(i) Limit cycles.

(ii) Jump resonance.

(iii) Deadzone. 6(CO1)